Benchmarking ImageNet Transfer Learning for Restricted Dog Image Classification

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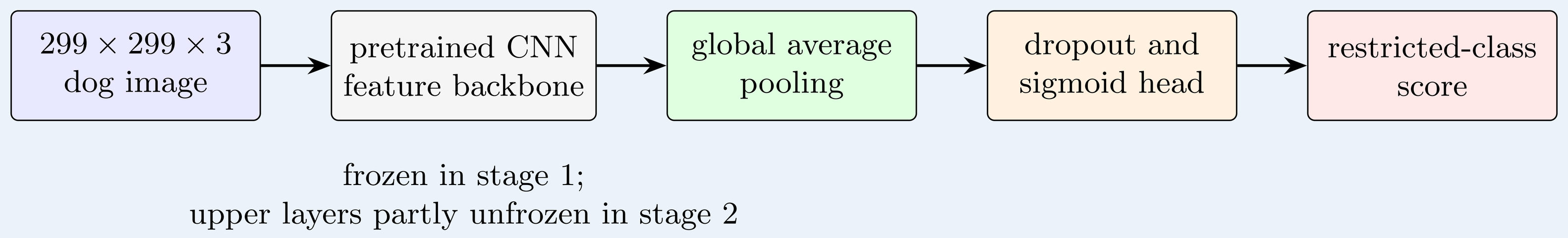
Research Aim

To benchmark six ImageNet-pretrained CNNs for restricted-versus-unrestricted dog-image classification and evaluate the effect of limited fine-tuning.

Dataset and Experimental Design

- Stanford Dogs image dataset: **30 breeds**.
- **6 restricted** and **24 unrestricted** breeds.
- Fixed training, validation and held-out test splits.
- Six ImageNet-pretrained CNNs benchmarked as frozen feature extractors.
- The selected model was fine-tuned by unfreezing its final 20 layers.
- Evaluation used accuracy, precision, recall, F1, ROC-AUC, PR-AUC, MCC and confusion counts.



Six Benchmark Architectures

VGG16 · ResNet50 · InceptionV3
Xception · InceptionResNetV2 · NASNetMobile

Reproducibility

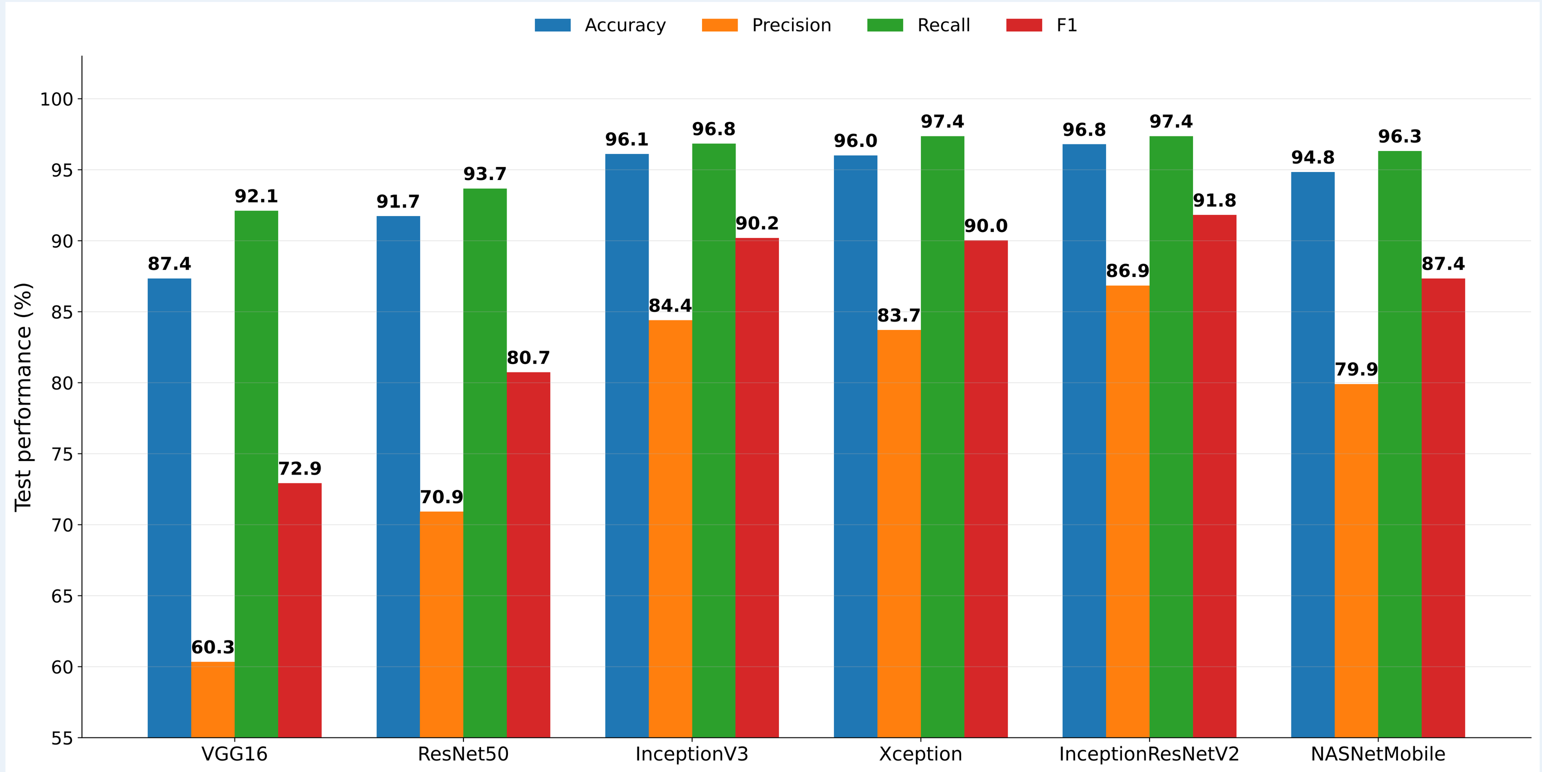
- Data preparation
- Frozen-model benchmark
- Selected-model fine-tuning
- Grad-CAM explanation

The archive retains the four notebooks, figures, result tables, predictions, checkpoints and complete LaTeX source.

Ethical Boundary

High predictive performance does not justify autonomous enforcement.

Frozen Model Benchmark



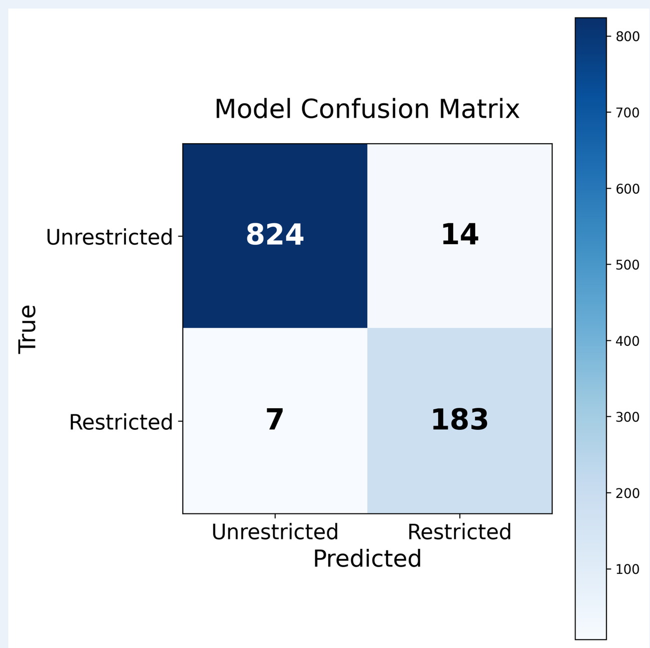
Best frozen model: InceptionResNetV2
97.08% accuracy · 98.42% recall · 92.57% F1

Fine-Tuning Result

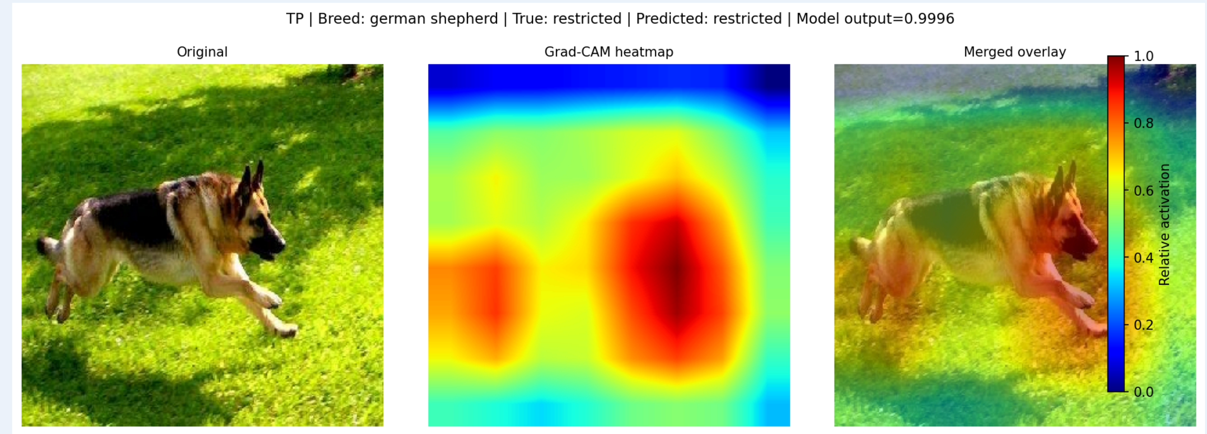
Metric	Frozen	Fine-tuned
Accuracy	97.08%	**98.25%**
Precision	87.38%	**93.88%**
Recall	**98.42%**	96.84%
F1	92.57%	**95.34%**
MCC	0.9101	**0.9428**

Key trade-off: 15 fewer false positives, but **3 additional false negatives**.
Fine-tuning improved overall balance and precision, but slightly reduced recall.

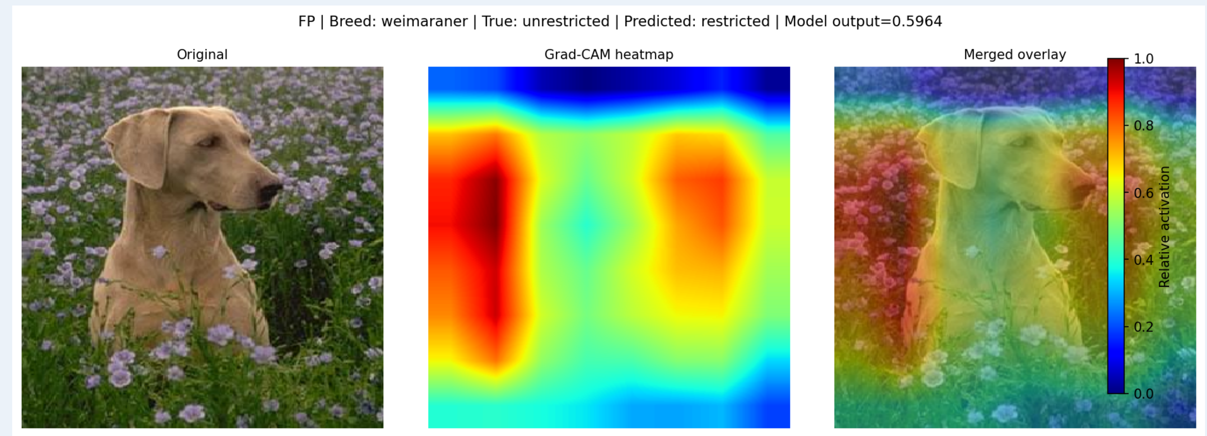
Fine-Tuned Confusion Matrix



Model Explainability



True positive: German Shepherd
Attention concentrated around the head and ears, which contributed strongly to the restricted prediction.



False positive: Weimaraner
Body shape, pose, coat contrast and contextual structure may have contributed to the incorrect restricted prediction.

Interpretive boundary: Grad-CAM heatmaps are post-hoc visual diagnostics. They indicate regions associated with a prediction, but they do not provide causal, behavioural or legal explanations.

Conclusions

- Six frozen ImageNet CNNs were benchmarked under one pipeline.
- InceptionResNetV2 produced the strongest frozen result.
- Fine-tuning improved precision and overall balance but reduced recall.
- The classifier cannot establish legal identity, ancestry, behaviour or dangerousness.